

Ectasia, acute hydrops and lamellar rehabilitation

Part 2 | Separate progression from poor optical quality

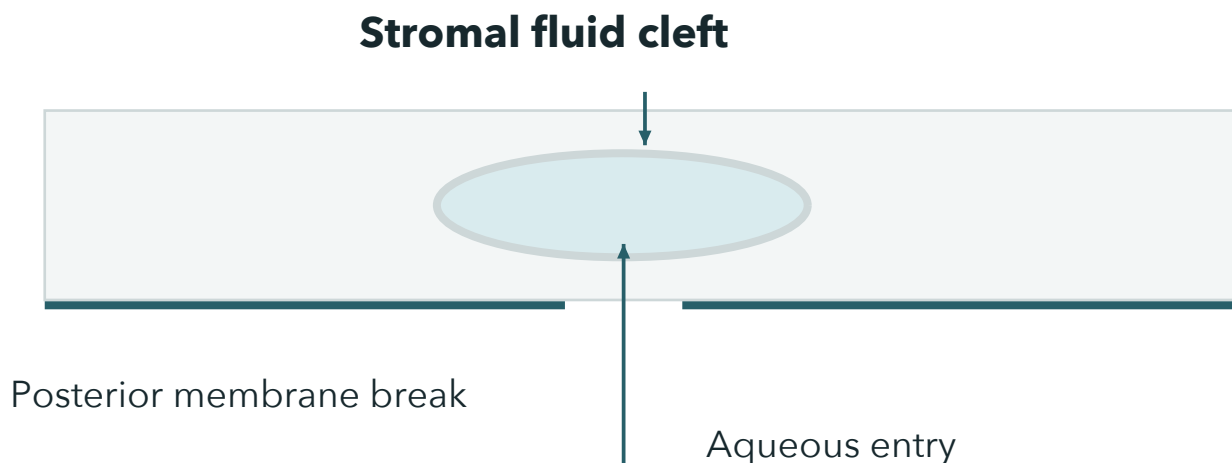
Stabilisation and rehabilitation answer different questions.

Problem	Assessment direction
Serial shape progression	Evaluate ectatic change with repeatable imaging
Irregular but stable optics	Assess spectacles or contact-lens rehabilitation
Dense stromal scar	Consider layer-appropriate optical replacement
Endothelial dysfunction	A different posterior-layer problem

Tomography, refraction and vision describe related but distinct aspects of ectasia. Measurement noise and lens warpage can mimic change. Cross-linking seeks to slow progression; it does not promise removal of a scar or correction of all higher-order aberrations. Assess stability before planning an irreversible refractive intervention.

Hydrops starts at the posterior barrier

A previously thin cornea can swell when aqueous enters the stroma.



Original section: a posterior membrane break allows aqueous into stromal clefts.

Acute edema can obscure the underlying ectatic geometry. AS-OCT helps display the membrane discontinuity and clefts. Treatment aims include comfort, epithelial protection and monitoring for complications. The swollen thickness is not evidence that the ectasia has structurally reversed, and scarring after recovery can change later rehabilitation.

Gas evidence needs endpoint discipline

Faster recovery does not establish superior final acuity.

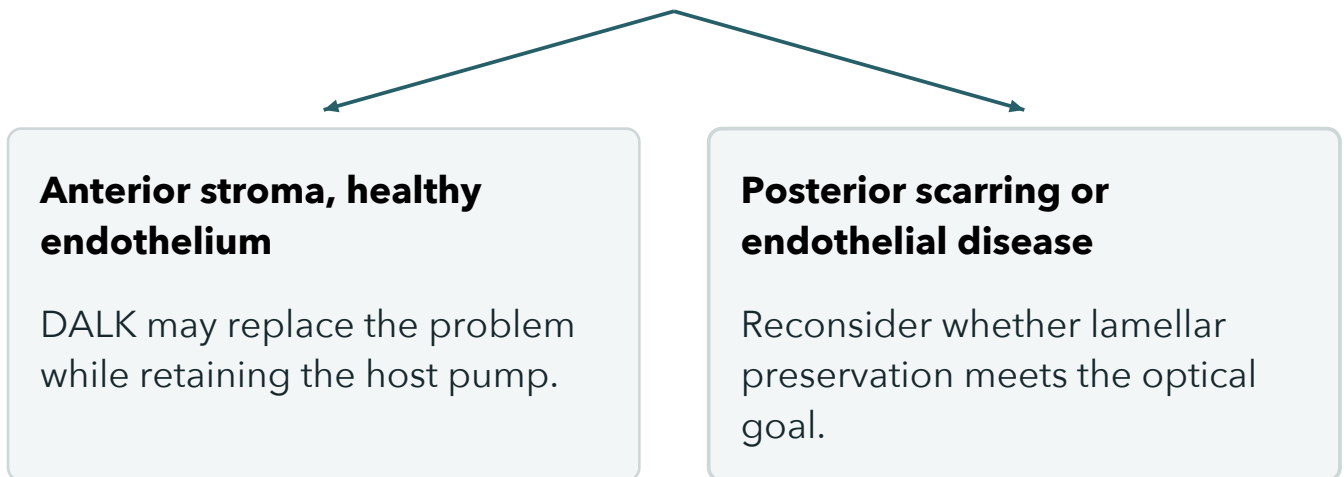
Reported result	Supported inference
Shorter edema-resolution time	Recovery may be accelerated
No final-acuity difference detected	Do not promise better final vision
Pupillary-block events	A procedure-specific risk needs discussion
Retrospective comparison	Residual confounding remains possible

The comparative C3F8 series is useful for counseling about time to recovery but cannot give randomized causal certainty. Consider defect geometry, patient burden and procedural risk. A treatment can improve one outcome without proving benefit in another; absence of a detected acuity difference is not proof of zero biological effect.

DALK preserves a healthy endothelial system

The posterior interface can limit what is technically achievable.

Where is the visually significant disease?

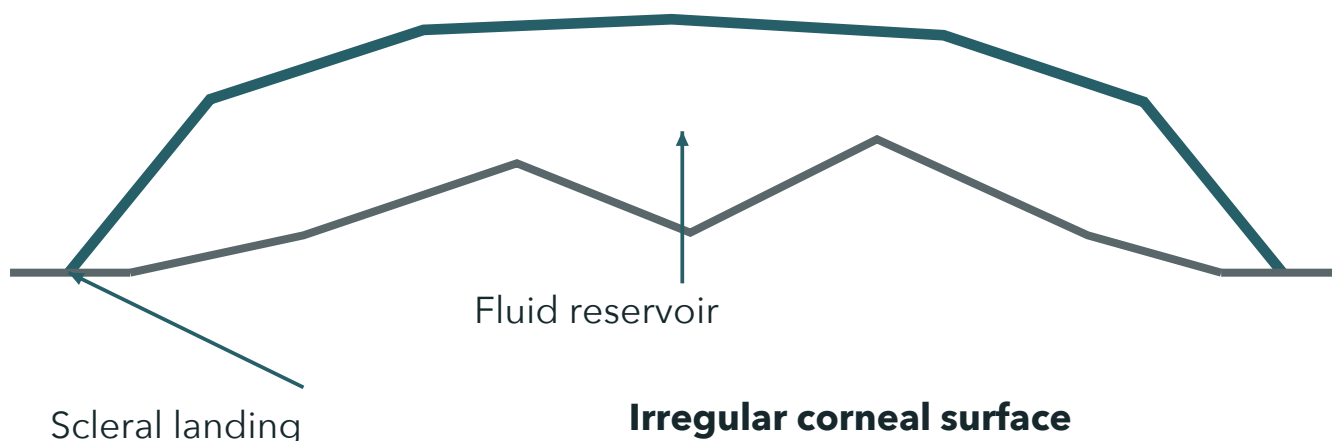


DALK can reduce endothelial rejection risk by preserving recipient endothelium, but it introduces a stromal or Descemet interface and has technical risks. A membrane perforation may alter the plan. Choose the operation from layer involvement and the realistic optical endpoint, rather than from the keratoconus label alone.

Contact lenses can rehabilitate irregular optics

A tear reservoir masks shape but does not eliminate surface risk.

Rigid lens vault



Original section: a scleral lens vaults the irregular cornea over a fluid reservoir.

Rigid and scleral lenses can improve irregular corneal optics without removing the ectasia. Fit, handling, hygiene, clearance and oxygen delivery matter. Persistent redness, discomfort or reduced vision warrants review rather than simply increasing wear time. The lens strategy and disease-stabilisation strategy should be discussed separately.